

| <b><u>Name</u></b>        | <b><u>Contribution (%)</u></b> | <b><u>Work</u></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ntando Thato<br>Mazinyane | 20%                            | <ul style="list-style-type: none"> <li>• I was assigned to do the Use Cases section and to assist with the Goal model's section.</li> <li>• Conducted Use case research for Duolingo and drew up some use case diagrams and tables.</li> <li>• I introduced the use case section and did function requirements.</li> <li>• I set up the Opening page, table of contents and moved a few sections around in order and placed them correct order.</li> <li>• Cleaning the document up to get ready for submission.</li> <li>• Edited mistakes based on Use case feedback</li> <li>• I worked on the Domain Model</li> <li>• Added pictures to Rich pictures</li> </ul> |
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|               |     |                                                                                                                                                                                                                                                                                                                                                                           |
|---------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |     | <ul style="list-style-type: none"> <li>• Cleaned up some part of the documents along with others.</li> </ul>                                                                                                                                                                                                                                                              |
| Andrey Pricop | 15% | <ul style="list-style-type: none"> <li>• I was assigned to help with the stakeholders.</li> <li>• I created the onion model and the Duolingo Stakeholder Model.</li> <li>• Spelling and grammar check</li> <li>• I worked on the Domain Model.</li> <li>• Structure for the document.</li> </ul>                                                                          |
| Matthew Moran | 15% | <ul style="list-style-type: none"> <li>• I was assigned to create the System Vision.</li> <li>• I created the System Vision Diagram and description.</li> <li>• I proofread.</li> <li>• I worked on the Domain Model</li> </ul>                                                                                                                                           |
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| Jake George   | 15% | <ul style="list-style-type: none"> <li>• I created the UML overview diagram.</li> <li>• I created the 3 use case tables.</li> <li>• I assisted with writing the document.</li> <li>• I worked on the Non-Functional Requirements.</li> </ul>                                                                                                                              |

|  |  |                                                                                     |
|--|--|-------------------------------------------------------------------------------------|
|  |  | <ul style="list-style-type: none"><li>• I worked on sections 2.1 and 4.1.</li></ul> |
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**Software Requirements Specification**  
**Duolingo**

**Version: 2**

**23/11/2025**

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## 1. Introduction

### 1.1 Purpose

This document describes the system requirements for Duolingo, A language learning platform that offers free and paid lessons through mobile and web applications.

It explains the purpose, context, and main functions of the system.

It also outlines who the system is for, what goals it serves, and how these goals are organised.

The aim is to communicate a clear, shared understanding of the system between technical and non-technical audiences.

It supports future design, testing, and maintenance activities.

### 1.2 Scope, Intended Audience and Use

This document is intended for:

- Project stakeholder: Product owners, developers, designers, marketing, and teachers using Duolingo for schools.
- Technical staff: Software engineers, UX designers, and system analysts involved in implementation and updates.
- Decision makers: Executives and investors who evaluate how well the system aligns with Duolingo's mission and business strategy
- Students and end users: People interested in how Duolingo manages learning goals and system quality.

Each section of this document can be read independently.

For instance, stakeholders may focus on goals, while developers may focus on requirements and system models.

### 1.3 Definitions and Acronyms

| Term | Description                                                                        |
|------|------------------------------------------------------------------------------------|
| SRS  | Software Requirements Specification                                                |
| UX   | User Experience                                                                    |
| XP   | Experience Points used for gamified progress                                       |
| UI   | User Interface                                                                     |
| NFR  | Non – Functional Requirements                                                      |
| KAOS | A forma goal modelling approach used to define, refine, and structure system goals |

## 1.4 References

- [1] Duolingo Inc., "Official Website and Company Reports," 2025. [Online]. Available: <https://www.duolingo.com>. [Accessed: Oct. 15, 2025].
- [2] H. Lee, "System Analysis Lecture Notes (Week 1 – 8)," 2025.
- [3] *IEEE Standard 830-1998: Recommended Practice for Software Requirements Specifications*, IEEE, 1998.
- [4] B. Penzenstadler, "Requirements Engineering," California State University, Long Beach, 2014.
- [5] I. Alexander and S. Robertson, "Understanding Project Sociology by Modelling Stakeholders," *IEEE Software*, vol. 21, no. 1, pp. 71–77, Jan. 2004.

## 1.5 System Overview

Duolingo is a cross-platform system that delivers language learning through gamified lessons, adaptive feedback, and data-driven recommendation. The system connects millions of learners with personalised content using an algorithm that adjusts to user performance.

Main features include:

- Lesson modules, exercises, and progress tracking
- Points on rewards (XP, streaks, leagues)
- Notifications and reminders to encourage consistency
- Duolingo for schools which helps teachers track students' progress
- Premium subscription (Super Duolingo) that removes ads and adds learning insights.

While the system is educational, it also operates as a business platform that balances learning outcomes with revenue through ads and subscriptions. This creates an important link between learning effectiveness and commercial goals, which will appear later in the Goal Model,

## 2. Overall Description

### 2.1 Product Perspective

Duolingo is a **multi-platform language-learning system** available on mobile, web, and tablet devices. It operates as an integrated ecosystem that combines **learning content, user engagement tools, and data analytics**. The system relies on cloud-based infrastructure to store user profiles, lesson data, and performance statistics.

This section delivers a full overview of the Duolingo language learning system that outlines the purposes, users, and context overall. We explain how the system integrates well with the language education domain and identify the relevant stakeholders who interact with it. The goal is to give a clean explanation of how it works and factors influencing its popularity today before giving the specific requirements in section 3.

From a broader view, Duolingo interacts with several external services:



- **App stores** for distribution and updates.
- **Payment system** for Super Duolingo subscriptions.
- **Ad Network** for revenue in the free version.
- **Social Media API's** for sharing progress and achievements.

While the platform is mainly educational, it functions as a **business system** that connects teaching, gamification, and marketing.

This balance between learning and engagement defines how the system's goals are shaped.

*Figure 2.1.1 - Stakeholder Matrix*

| Stakeholder                   | Role           | Functions      | Motivation                                                             | Main Concern                                | Availability | Relations          |
|-------------------------------|----------------|----------------|------------------------------------------------------------------------|---------------------------------------------|--------------|--------------------|
| Free Learners                 | User           | End-user       | Learn a language for free desire for progress and gamified experience. | Potential ads disrupting the service speed. | High         | Ads, Notifications |
| Paid Learners                 | User           | End-user       | Access ad-free, offline, and progress features.                        | Billing can be unreliable                   | High         | Billing, Support   |
| Parents /Guardians            | Owner /Partner | Decision maker | Ensure child safety, parental controls, and privacy                    | Misuse or exposure                          | Medium       | Privacy            |
| Teachers                      | Partner        | Info provider  | Assign lessons, track student progress                                 | Poor classroom integration                  | Medium       | Classroom          |
| Classroom Students            | User           | End-user       | Complete assigned lessons and show progress                            | Access limits                               | Medium       | Accounts           |
| DET Candidates                | User           | End-user       | Take certified language test online                                    | Fairness and technical issues               | Medium       | Identity,          |
| Content Designers /Linguists  | Developer      | Implementer    | Ensuring lesson quality, linguistic accuracy, and coverage             | Scalability                                 | High         | Curriculum         |
| Mobile Engineers              | Developer      | Implementer    | Build and maintain app stability, UI, and performance                  | Crashes                                     | High         | CI/CD, SDKs        |
| ML/ Personalisation Engineers | Developer      | Implementer    | Deliver accurate, adaptive recommendations                             | Bias and data limits                        | High         | Data Platform      |
| Data Scientists /Analysts     | Developer      | Info provider  | Provide insights and A/B testing analysis                              | Data reliability                            | High         | Analytics, Product |
| UX Researchers/ Designers     | Developer      | Info provider  | Improve usability and accessibility                                    | Inclusivity                                 | Medium       | Users, Ally        |
| Community Moderators          | Support        | Support        | Monitor community discussions                                          | Conduct and content                         | Medium       | Policies           |
| Customer Support              | Support        | Support        | Assist users with issues or refunds                                    | Response time                               | Medium       | Billing            |

|                                          |                     |                |                                               |                             |        |                    |
|------------------------------------------|---------------------|----------------|-----------------------------------------------|-----------------------------|--------|--------------------|
| Marketing                                | Owner               | Decision maker | Increase daily active users and conversion    | Notification backlash       | Medium | Notifications, Ads |
| Legal and compliance                     | Regulator /Internal | Regulatory     | Maintain GDPR/CCPA/CO PPA compliance          | Legal risks                 | Medium | DPO Product        |
| Data Protection Officer                  | Regulator           | Regulatory     | Oversee privacy and DPIAs                     | Security Breaches           | Medium | Security           |
| App Stores (Google play, app store etc.) | Partner             | Decision Maker | Ensuring compliance with the general policies | Violations of policies      | Low    | Billing, Releases  |
| Payment Processors                       | Partner             | Implementer    | Process in-app purchases securely             | Transaction failures        | Low    | Billing            |
| Cloud Provider                           | Partner             | Implementer    | Ensuring there is uptime and scalability      | Costs and uptime capability | Low    | SRE /Dev Ops       |
| Consumer Protection Regulators           | Regulator           | Regulatory     | Prevent dark-patterns and exploitative design | User wellbeing              | Low    | Legal              |

2.1.2 Stakeholder Matrix Analysis

This stakeholder matrix identifies the key personnel involved with the language learning system Duolingo which outlines their roles, responsibilities and motives.

- Primary stakeholders such as free learners, paid learners and DET candidates prioritise effectiveness of fair and accessible learning, having concerns about ads, billing and reliability.
- Secondary stakeholders such teachers, parents and students have a fixation on privacy, safety having integration through their goals.
- Developers like engineers, and UX designers guarantee scalability, functionality and excellent quality of the system.
- External partners like app stores and payment processors maintain compliance and the general framework.

Overall, a successful system requires a good set of valuable stakeholders from different departments to sustain the general upbringing of Duolingo.

Figure 2.1.2 - Onion Model

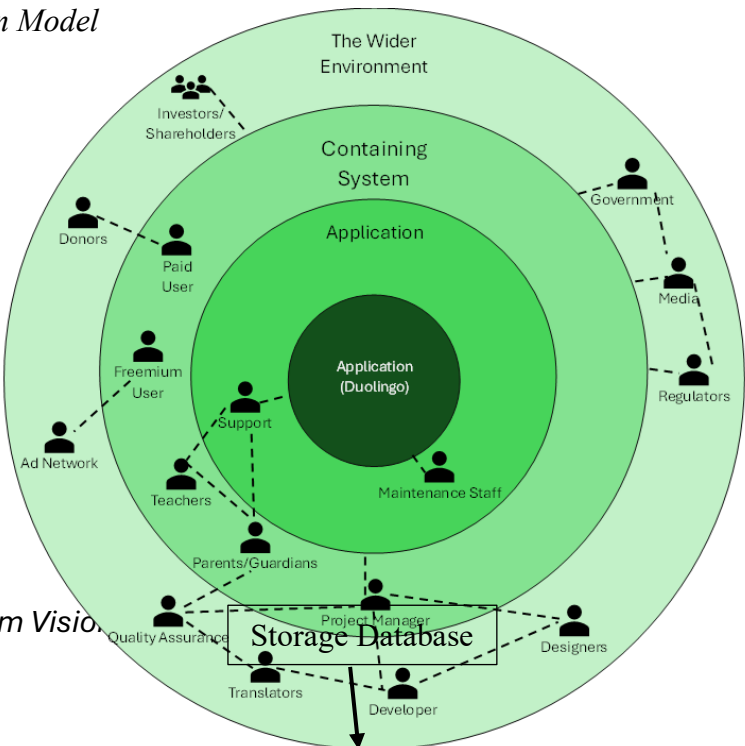
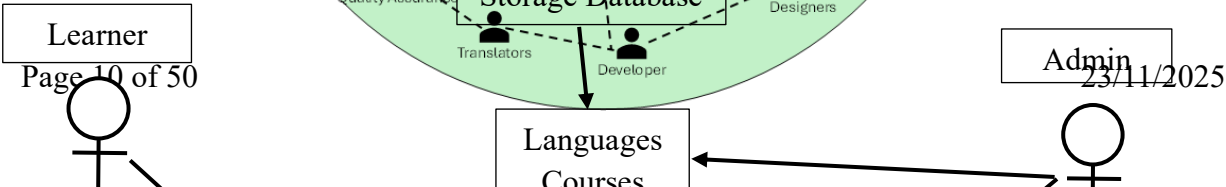
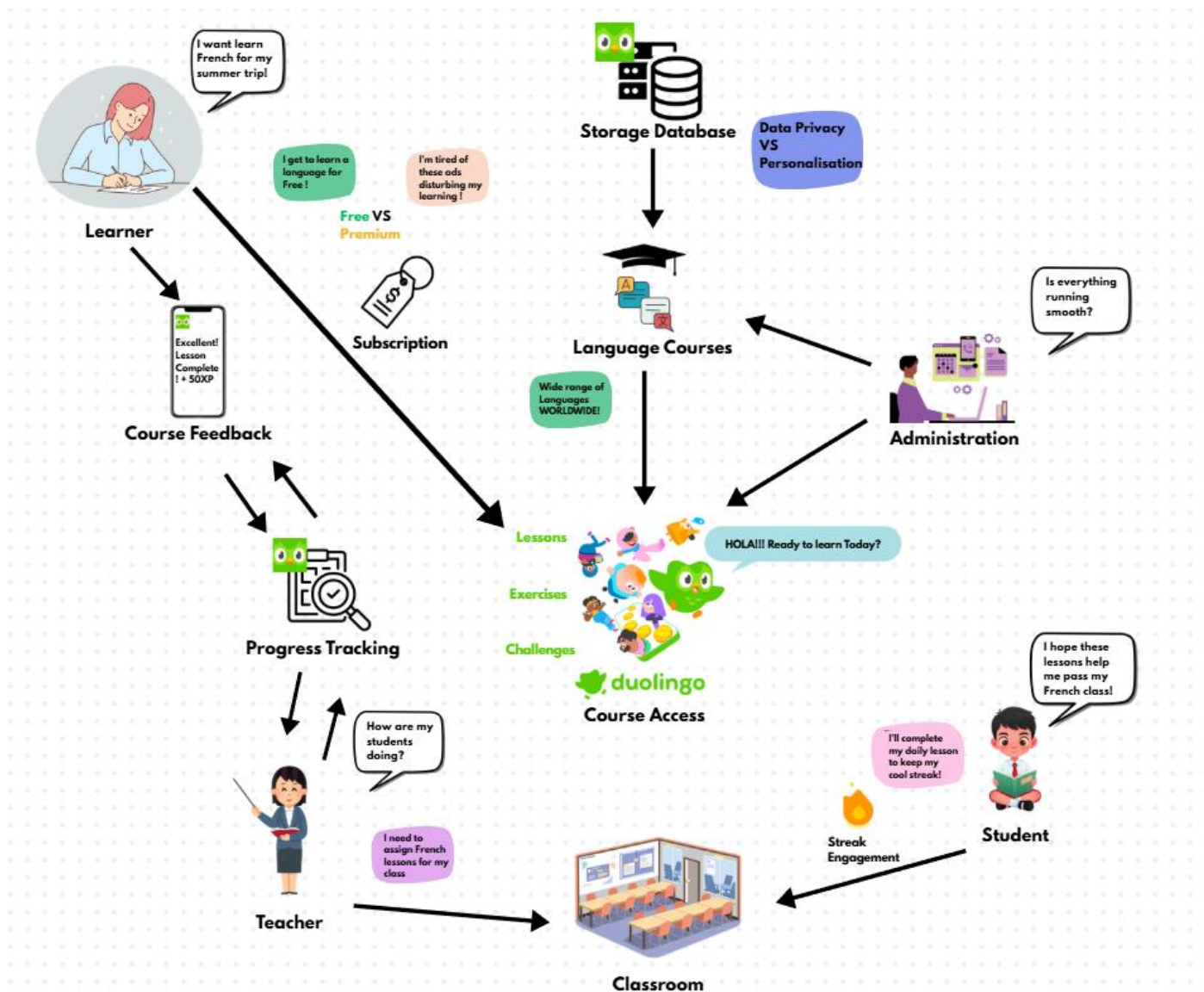


Figure 2.1.3 -System Vision







S

Above is the system vision diagram. It starts with the storage databases for the company providing access to language courses via an API. Users (Learner, students and teachers) create an account which allows the use of the API to stream the language course information to their app. The first user, learner, is your average person. They can buy a subscription (or not) as we use the freemium model giving both a free with ads or paid, uninterrupted experience. Each language course is made up of lessons and exercises which vary in difficulty and topic. These courses (and all of their content) as well as the database servers themselves are updated and given upkeep by the administrators/moderators. These administrators are either language experts in their chosen language who update the language courses by providing better grammar or new lessons about differing topics in today's world. Our second user, the student, is a language student who is in a class of other students. The teacher user can create an inclusive e-classroom to provide lessons to do and also receive feedback based on how any one student performs on said lessons. The students access the language course through the classroom portal which is set up by the teacher and hosted by Duolingo. The classroom access requires a subscription made by the teacher / school setting up this feature. User 1, the learner, can also avail of our progress tracker and course feedback. The depth of such features depend on the subscription held by the learner, however, the feedback can still help users along their language journey regardless of subscription. Progress tracking also takes place in the form of the "daily streak", which, the

longer it is held, users can avail of better rewards in the form of XP, which can unlock different new areas of the language course.

## 2.2 Product Functions

- Duolingo offers language lesson services through its quick and snappy interactive lessons and engagement features.
- It is delivered in structured lessons and adaptive exercises.
- Duolingo provides reminders and notifications to encourage daily use.
- It includes flexible reading, listening and speaking exercises that aim to boost the user's capability of learning that language.
- Duolingo provides a unique streak system which encourages loyalty within their userbase allowing them to come back each day to do a specific learning exercise to maximise their potential and potentially purchase subscription software displayed.
- Duolingo also tracks your progress using XP within each exercise to maximise users' competitive nature to continue doing exercises, with a leaderboard included against other users to review your progress.
- And as profit it also enables users to subscribe to premium features,
- Overall, Duolingo's features provide a background of diverse types of users such as education, entertainment, and people needing general accessibility to learn a language.

### 2.2.1 Purpose of Goal Modelling

I used Goal modelling to identify and organise what Duolingo aims to achieve from different perspectives like business, usage, and system.

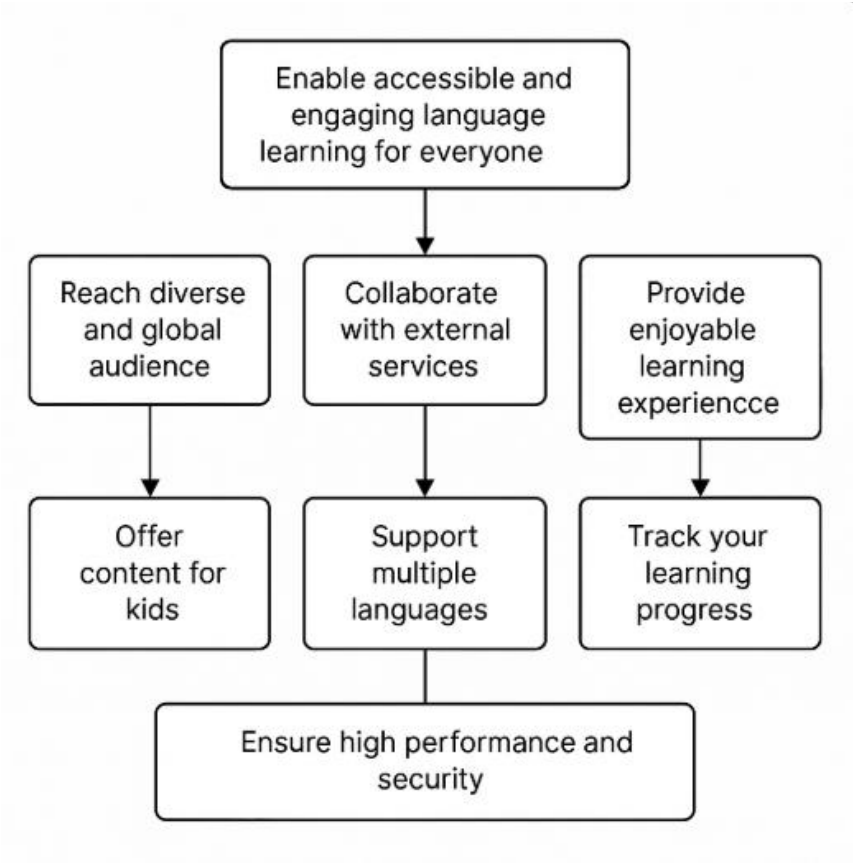
This model helps clarify *why* the system exists, *what* it does, and *how* it should perform and what conflicts with each other.

The structure combines **KAOS notation** (Achieve, Maintain, Avoid) with a simple hierarchical format to make it easier to read.

It shows how Duolingo's purpose breaks down into practical, measurable goals that involve learners, teachers, parents, and partners

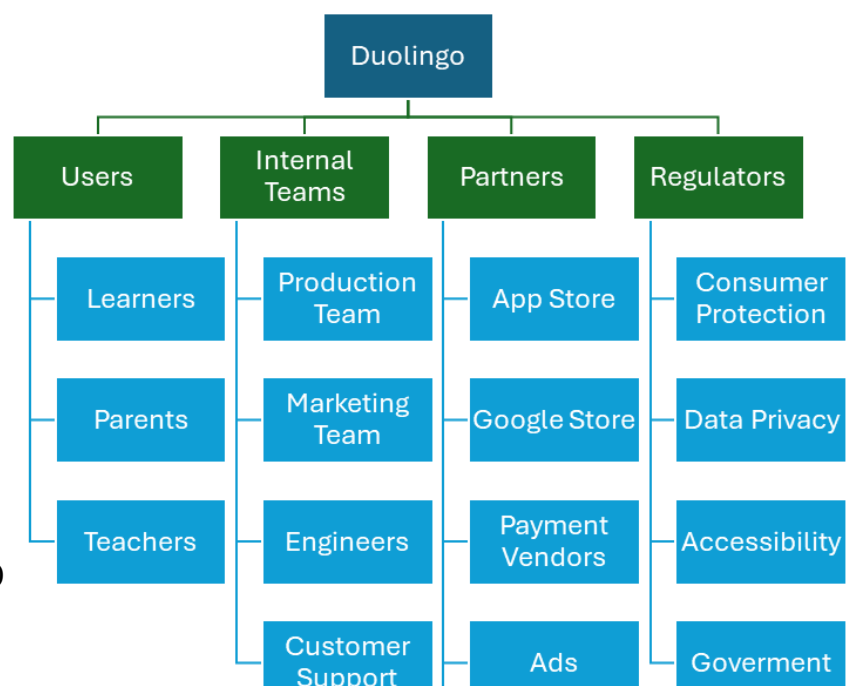
In the diagram, each goal connects upward toward the main system purpose "Enable accessible and engaging language learning for everyone and easier collaboration."

Figure 2.2.1.1 - Simplified Hierarchy Diagram



Above diagram is not a Goal Model Diagram. It is hierarchy diagram which gives an overall simplified vision of a Goal Model Diagram in hierarchical form. A bird’s eye vision in simplified form to grasp the concept in first glance.

Figure 2.2.1.3 - Stakeholder Structure



### 2.2.2 Main Goal

This Goal covers 3 ideas:

- **Accessibility:** Anyone can learn the language regardless of cost or background and anywhere
- **Engagement:** Users stay motivated through fun, short, daily interactions.
- **Safety:** Protect younger learners through privacy and parental oversight.
- **Collaboration:** Work with educators, parents and other platform to extend learning beyond Duolingo.

### 2.2.3 Business Goals (Why it's built)

#### 1. Expand global language accessibility

- Keep lessons free for all users while offering optional upgrades.
- Stakeholders: Executives, NGOs, education partners.

#### 2. Create Sustainable business and partnerships

- Combine ads, premium subscriptions, and partnerships with YouTube or educational publishers
- Stakeholders: Executives, marketing and external partners.

#### 3. Promote daily user engagement

- Use streaks, notifications, and achievements to encourage regular practice.
- Stakeholders: Product managers, UX designers.

#### 4. Support Collaboration with educators and institutions.

- Provide teaching tools, shared dashboards, and school level features.
- Stakeholders: Teachers, administrators, learners.

#### 5. Enhance brand trust and visibility

- Build a positive comment inclusive image for families, schools, and partners.
- Stakeholders: Marketing, PR, support team.

#### 6. Expand into child friendly learning markets

- Create a safe, fun environment for children aged 6 – 12 using playful, guided learning.
- Stakeholder: Parents, schools, educational partners and product managers.

#### 7. Collaborate with educational and media partners.

- Integrate verified content from sources like **YouTube Kids** or **Khan Academy** or **PBS Kids**.
- Stakeholders: External partners, content developers, educators, teachers.

#### **2.2.4. Usage Goals (What it does)**

##### **1. Deliver adaptive, personalized lessons**

- Tailor exercises to skill level and learning pace.
- Stakeholders: Learners, UX designers, data scientists.

##### **2. Create Sustainable business and partnerships**

- Have business partnerships and collaborations which enhances brand recognition.
- Stakeholders: Marketing team, educators, business partners, influencers

##### **3. Integrate External media and learning content**

- Recommend short videos, songs, or podcasts for immersive learning.
- Stakeholders: Content team, partners, learners.

##### **4. Streaks and XP Leaderboards**

- Encourage the users to use the software more with competence and increases the usage of the software which promotes learning.
- Stakeholders: Learner, Teachers, Educators, Parents and any form of targeted users.

##### **5. Support peer learning and team challenges**

- Encourage friendly competition and teamwork.
- Stakeholders: Learners, moderators.

##### **6. Enable teacher-student collaboration**

- Teachers create groups, assign lessons, and monitor results.
- Stakeholders: Teachers, students, schools.

##### **7. Provide detailed progress dashboards**

- Summarise learning progress for users, teachers, and parents and adjust screen time.
- Stakeholders: Learners, educators, parents.

##### **8. Enable parental supervision tools**

- Parents can track usage time, set limits, and review results.
- Stakeholders: Parents, developers, support staff.

##### **9. Offer a child learning mode**

- Simplified design, safety locks, and guided lessons for kids.
- Stakeholders: Parents, teachers, UX designers, young learners.

##### **10. External Media and learning content integration**

- Used for the media usage to integrate the third party which makes the user access vary or different types of method. For e.g., have YouTube videos underneath the language learning mode (e.g. pronunciation video)
- Stakeholders: Learner, parents, educators

##### **11. Educator Resource hub**

- Storage for resources for educators for easy access when teaching. Give control access for specific actions and analytics to track the students' performance.
- Stakeholders: Learners, Teachers, Educators, Parents.

#### **2.2.5. General System Goals (How it performs)**

##### **1. Maintain high reliability and scalability**



- Handle millions of users and collaborative sessions smoothly.
- Stakeholders: Developers, system engineers.
- 2. Ensure privacy, data security, and compliance**
  - Follow GDPR and COPPA-K regulations for both adults and children (collect minimal data for children). Authentication and encryption data and secrets including handling/storing of data.
  - Stakeholders: Legal, developers, parents.
- 3. Optimise performance across devices**
  - Support low end phones and older browsers effectively.
  - Stakeholders: Developers, QA testers, users.
- 4. Enable secure third-party integrations**
  - Allow safe API connections with partners like YouTube Kids and schools.
  - Stakeholders: Developers, partners, IT teams.
- 5. Provide AI based analytics and content recommendations and filtering**
  - Use data to refine lessons and suggest relevant material.
  - Stakeholders: Data scientists, content designers, legal, external partners.
- 6. Implement strong child protection controls**
  - Restrict tracking, filter inappropriate content or external (YouTube) content for age-appropriate material, and anonymize data,
  - Stakeholders: Legal, developers, parents, external partners.
- 7. Support accessibility and assistive technologies**
  - Provide subtitles, voice recognition, and visual aid futures.
  - Stakeholders: Accessibility teams, learners with special needs.
- 8. Observability (logs and metrics)**
  - Logs and metrics which are stored in the database. Traces and alerts based on what is collected from the user. Used to improve the products and track if anything goes wrong.
  - Stakeholders: Data Scientist, Developers, Cloud operators, IT teams.
- 9. Disaster recovery and backups.**
  - Recovery team in case of a disaster or an error in the system due to many factors.
  - Stakeholders: Cloud Operators, , IT teams, Software Engineers, Data Scientist, Developers.

**These are general system goals that applies for the whole of usage goals in general as these must be met for the software to be handled in general with no issues/least issues as possible. The more detailed version has been implemented in the Goal Model which I have pasted below and also in a PDF Format.**

### **Strengthen Collaboration Aspect**

This displays the collaboration between the goals

- **Cross-Platform Integration:**  
Duolingo can link with platforms like YouTube Kids, Khan Academy, or school LMS systems to enrich learning material.
- **Teacher + Parent Collaboration:**  
Teachers and parents can jointly monitor progress and recommend activities.

- **Corporate or NGO Collaboration:**

Work with UNICEF, UNESCO, or national language programs to support free access for children in developing regions.

- **Social Collaboration:**

Allow community-created stories, flashcards, or games reviewed by educators — encouraging participatory learning.

## 2.2.6. Goal Relationships and Conflicts

### Supportive links:

- Child learning mode supports expand child friendly markets and enhance brand trust.
- Integrate external media supports collaborate with educational partners.
- Parental supervision tools support safety and trust.
- Reliable infrastructure supports collaboration and partnerships.
- AI analytics supports engagement and personalization.

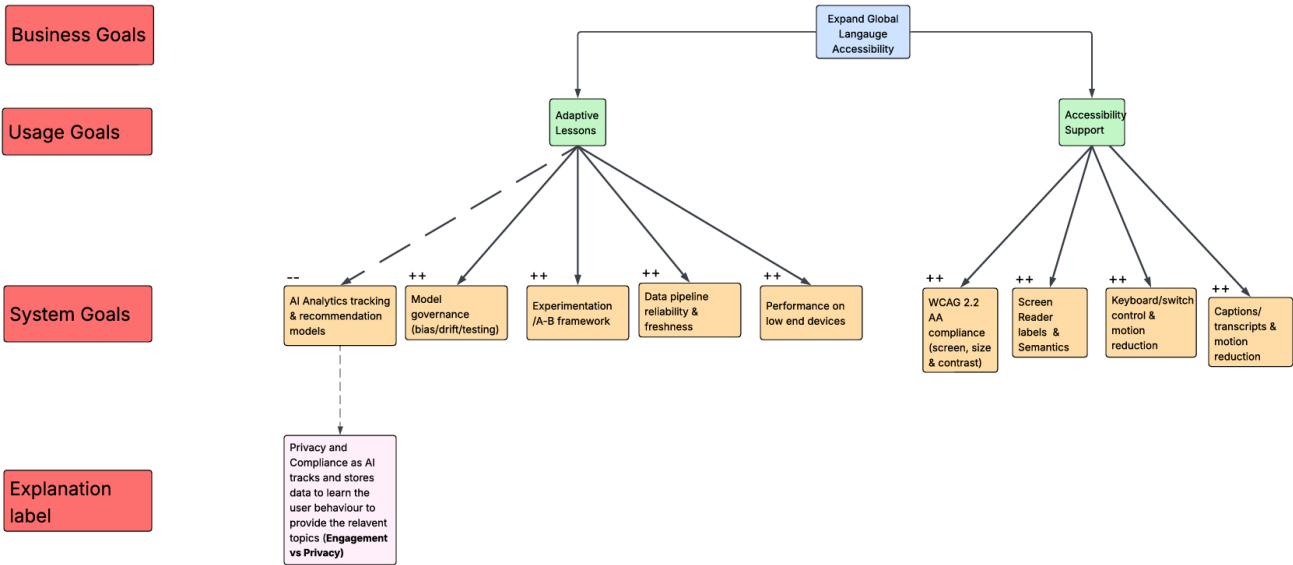
### Conflicts:

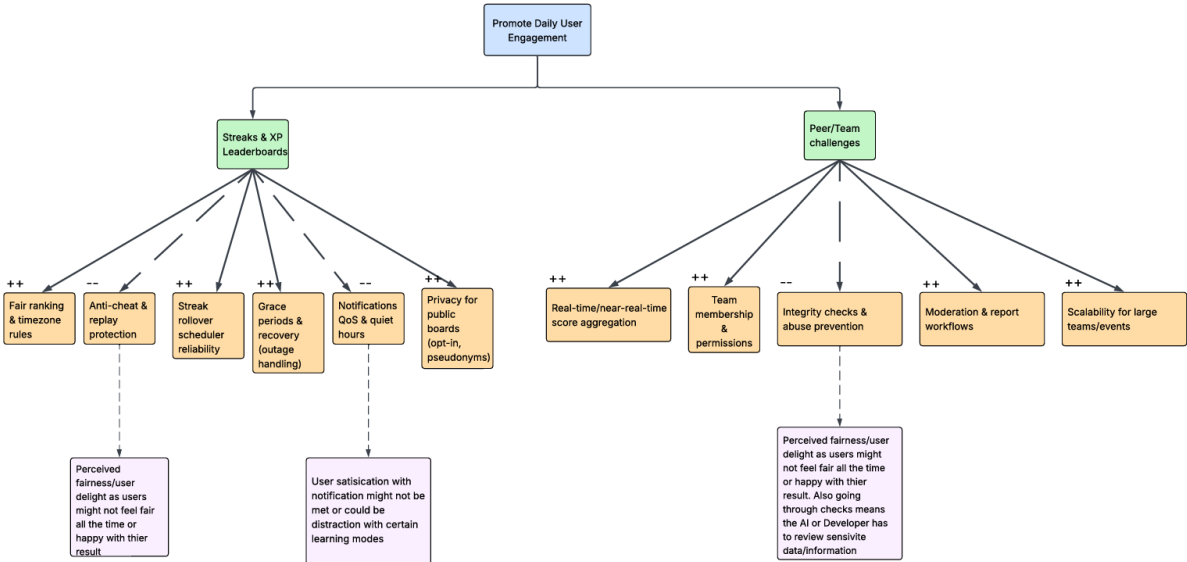
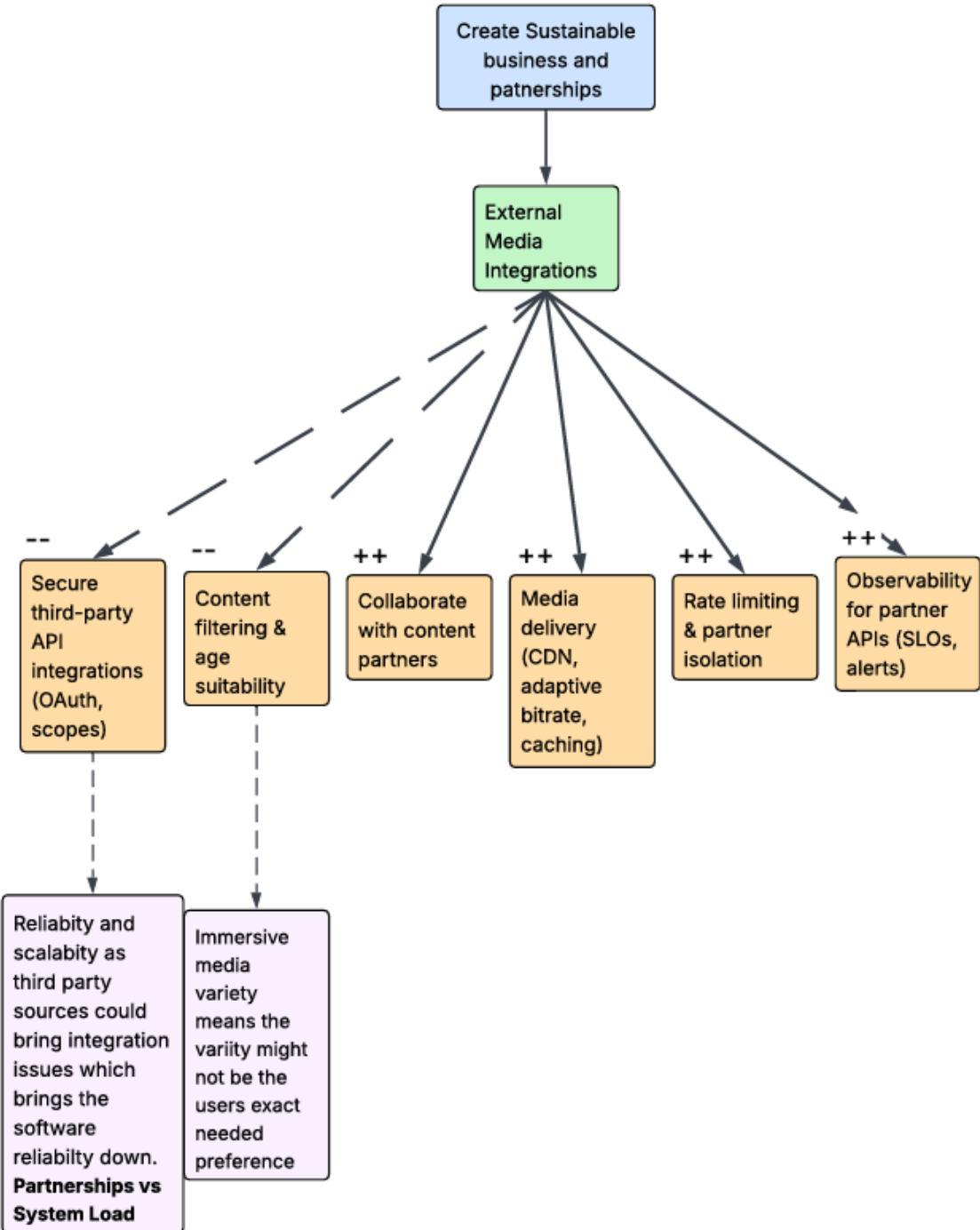
1. **Engagement vs Privacy:** Personalization relies on user data, which may risk privacy, especially for children
2. **Partnerships vs System Load:** Integration with You Tube and schools increase data flow and system stress.
3. **Child Mode vs Flexibility:** Simplified interfaces may limit access to advanced learning tools.

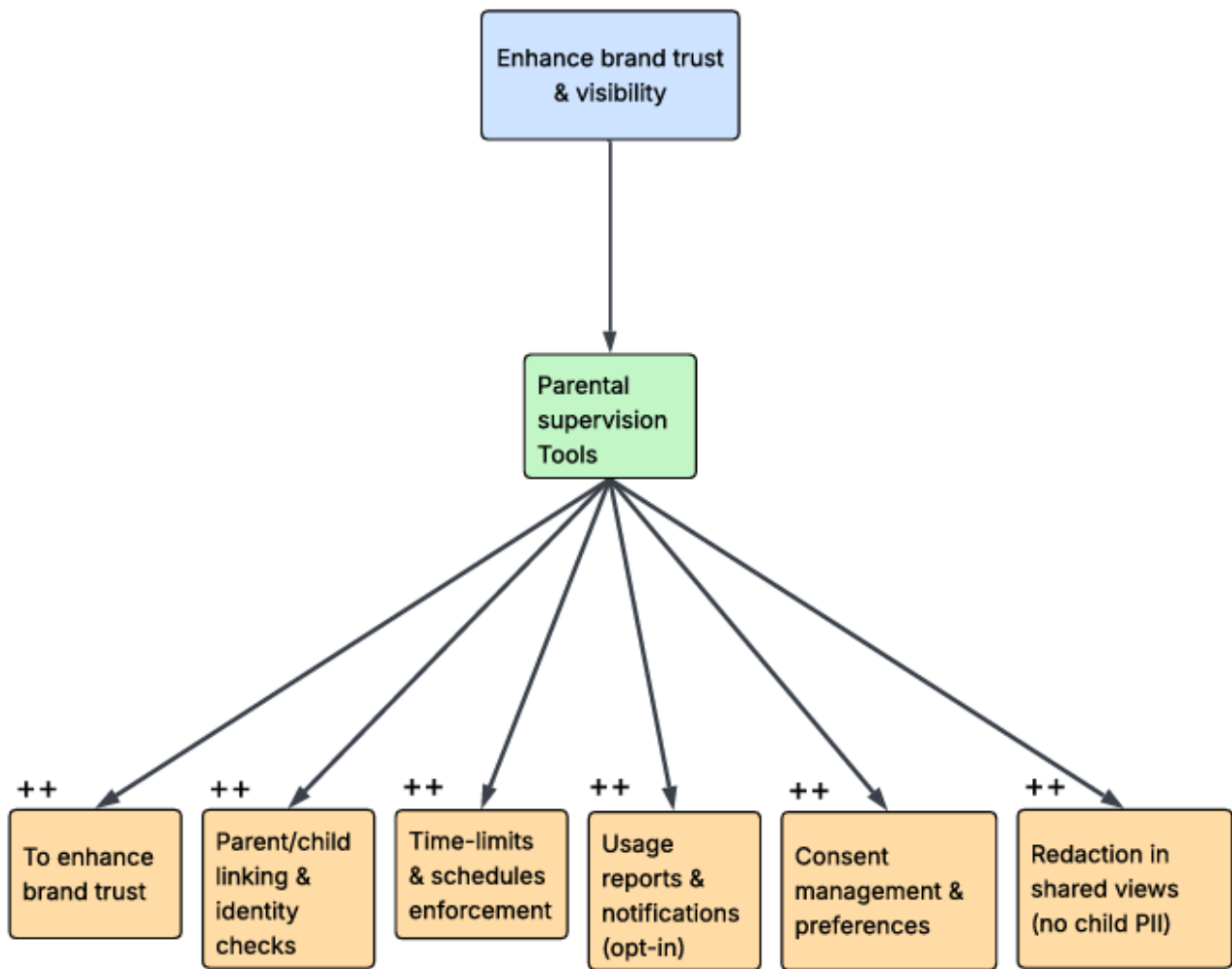
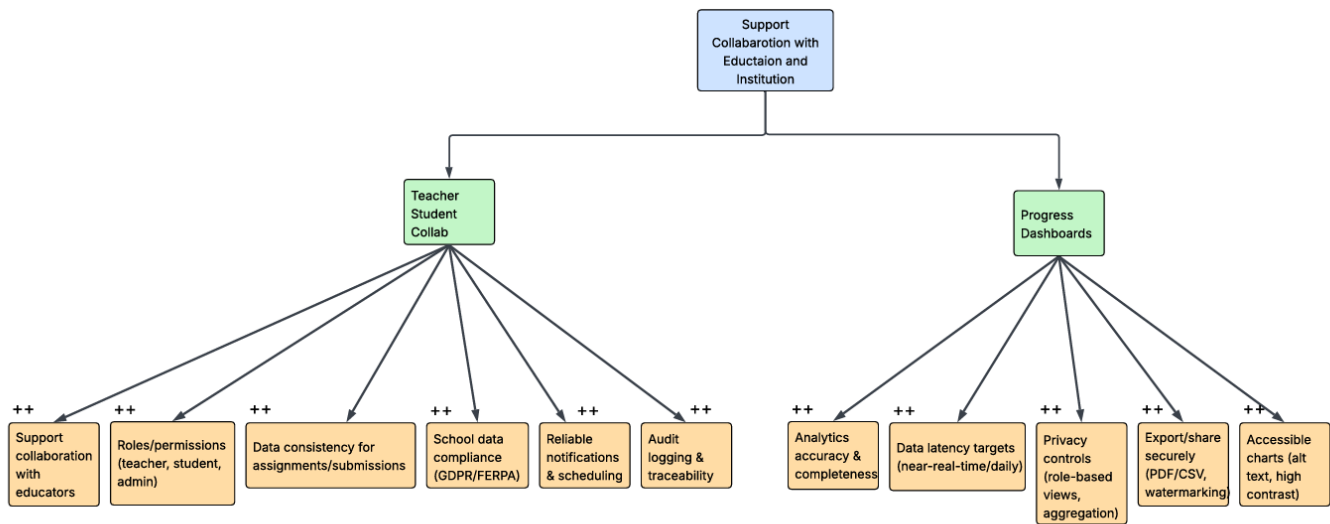
These conflicts help identify design trade-offs between growth, safety, and performance which are key aspects of system design decisions.

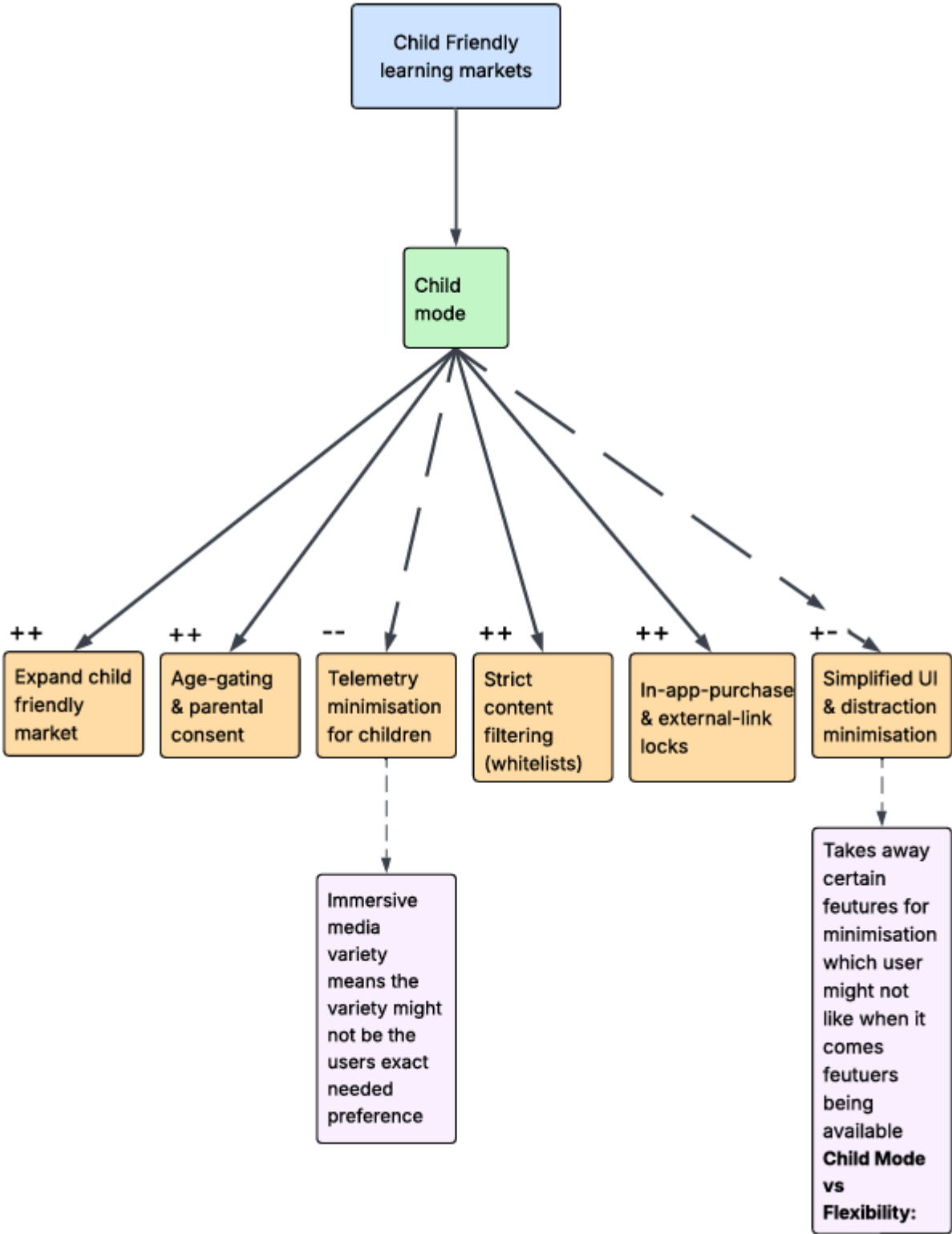
## 2.2.7. Goal Hierarchy Diagrams

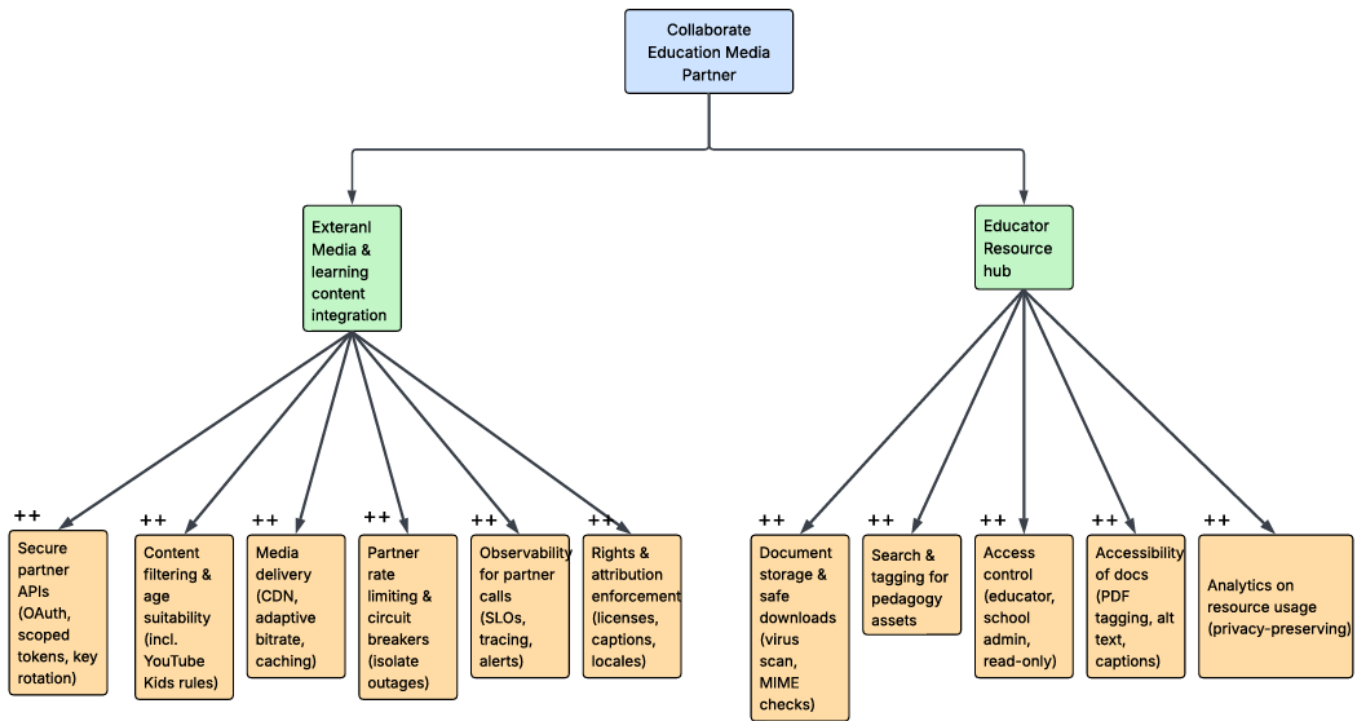
Figure 2.2.7.1 - Goal Model – *Goal Model also has been pasted as separate document in PDF Format*











## Colour Label Meaning for Goal Model

**Grey** = Heading Label

**Green** = Usage Goal

**Yellow** = System Goals

**Purple** = Notes for the conflicts

**Red Colour** = Labels

The above is the goals model diagram that I have uploaded separately under each section as it was not possible to fit the whole goal model in one section, so I split them into sections. Additionally, the whole diagramming in one page has been put as a pdf file.

## Legend:

- Solid arrows “++” == positive contribution
- Dashed arrows “- - -” = conflicts

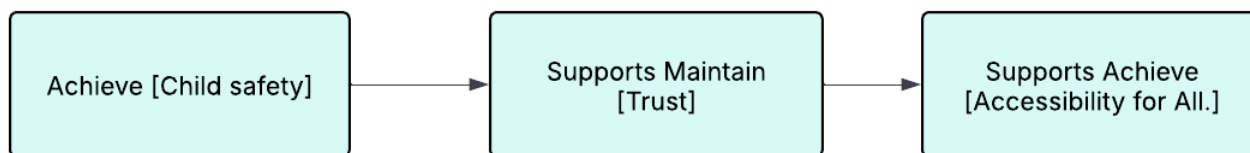
## Example:

- AI Analytics ++ Personalisation
- Child Mode – Advanced feature flexibility
- Engagement ++ Business growth
- Privacy – Engagement tracking

### 2.2.8 How to Read the Model

- Read from top to bottom – each level refines the one above.
- Arrows indicate mark conflicts or trade off.
- Goals are traceable, each one tied back to a stakeholder and a measurable outcome.

#### Example trace:



**Figure 2.2.9 - Stakeholder Goal Mapping**

| Stakeholder                                           | Related Goals                                                                     | Main Interest                                              |
|-------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Learners</b>                                       | Adaptive lessons, gamification, privacy, personalised lessons                     | Learn effectively, efficiently and stay engaged            |
| <b>Children/Young Learners</b>                        | Child mode, content filtering accessibility                                       | Safe, fun learning experience                              |
| <b>Parents / Guardians</b>                            | Parental tools, privacy, child safety                                             | Monitor progress and protect data                          |
| <b>Teachers / Schools</b>                             | Collaboration, dashboards, reporting, track class performance, manage assignments | Guide and track student learning, support student progress |
| <b>Developers / Engineers</b>                         | Reliability, Integration, Optimize Performance                                    | Stable and scalable systems and ensure smooth operation    |
| <b>Executives / Investors</b>                         | Partnerships, engagements, expansion, Revenue, Global reach                       | Long term growth and ensure sustainability                 |
| <b>Marketing / PR</b>                                 | Enhance brand trust reputation, engagement, outreach                              | Broader reach and user loyalty and grow user base          |
| <b>Legal / Compliance</b>                             | Privacy, child data laws, Protect user privacy                                    | Legal protection, compliance and trust                     |
| <b>External Partners (YouTube kids, Khan Academy)</b> | Integration, shared media                                                         | Collaboration and audience reach                           |
| <b>NGOs / Education Authorities</b>                   | Accessibility, child learning                                                     | Support equal education opportunities                      |
| <b>Data Scientist</b>                                 | Analytics and personalisation                                                     | Improve accuracy of learning models.                       |
| <b>Accessibility Teams</b>                            | Assistive features, inclusive design                                              | Ensure usability for all                                   |



## **2.3 User Characteristics**

The main user groups include:

- Individual learners: People learning new language for personal or professional reasons.
- Teachers and institution use: “Duolingo for Schools” to assign lessons/assignments related to a language and track students' progress.
- Developers and data scientists: Maintain the adaptive algorithms and user interface.
- Marketing and analytics teams: Monitor engagement, retention and conversion rates.
- Executives and investors: Oversee strategic direction and system growth.

Users vary in age, language proficiency, and learning goals.

Most learners access the platform via smartphone and expect short, intuitive interactions.

## **2.4 Constraints**

### **1. Global Data Protection Compliance**

The system shall comply with major privacy regulations such as GDPR (General Data Protection Regulation) for EU user rights, COPPA-K (Children’s Online Privacy Protection Act) for protecting those under 13 years of age, FERPA (Family Educational Rights and Privacy Act) for student data and relevant local data laws in all of Duolingo’s operating regions. This applies to all user data collected from learners, teachers, parents and educational partners.

### **2. Multi-Platform Support**

The system shall operate smoothly on Android 8 and newer, iOS 13 and newer, and modern web browsers such as Chrome, Safari, Firefox and Edge. The application shall provide reduced functionality for older devices and low-resource environments that would commonly be found in education settings to ensure language learning remains accessible across diverse socioeconomic contexts.

### **3. Network Dependence**

The system shall require persistent internet connectivity for cloud-synced features including progress synchronisation, media recommendations from partner platforms, real-time school assignments and parental monitoring tools. Offline functionality shall be restricted to pre-downloaded lesson content only ensuring core learning is still available while collaborative and fresh content features maintain their dependency on reliable network connections.

### **4. Third Party API Compliance**

The system shall adhere to external partner API requirements including rate limiting to prevent service overload, educational content filtering for age-appropriate materials, strict age verification protocols and restricted content categories for minor users. These compliance measures shall be enforced across integrated platforms such as YouTube Kids for media content, school Learning Management Systems (LMS) for educational data and secure payment gateways for subscription services.

### **5. Educational Content Standards**

The system shall maintain Duolingo's established linguistic standards including pronunciation accuracy, grammatical correctness and cultural appropriateness across all lesson content. The system shall enforce approved difficulty progression pathways to ensure concepts build upon previous knowledge maintaining consistent and age-appropriate educational journeys for teachers managing their classrooms, parents monitoring progress and school groups implementing standardised curriculums.

## **6. Advertising Restrictions**

The system shall prohibit advertisements in child mode, educational institution-managed accounts and regions with strict advertising regulations. This ad-free environment shall ensure safe, distraction-free and legally compliant learning spaces for minors, students and protected user groups across all operational markets.

### **2.5 Assumptions and Dependencies**

This project assumes:

- Users have access to smartphones or web browsers.
- The system continues to rely on Duolingo's proprietary AI and data analytics.
- Teachers and learners use the same underlying database for lesson progress.
- External payment and ad systems remain stable.

The system depends on continuous server uptime, stable APIs, and accurate language content from linguists and experts.

## **3. Specific Requirements**

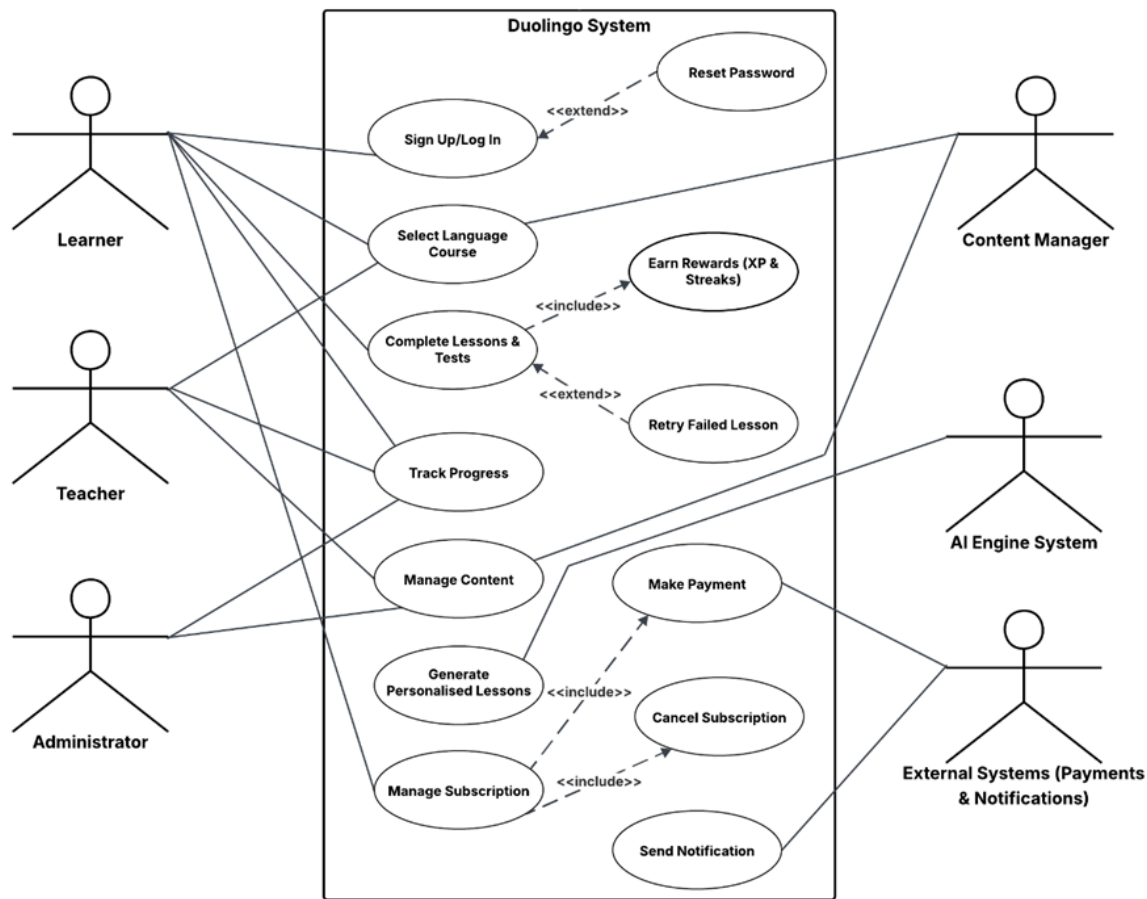
### **3.1 Introduction**

In this section, we will be diving into the specific functional requirements of the Duolingo System. This involves displaying how the system is expected to behave when users and supporting component interact with it.

To convey this, we utilized Use Case Models to set up scenarios between primary actors and their interactions. Functional requirements contribute by establishing specifications of what the system shall do in response to inputs.

### **3.2 Use Cases**

*Figure 3.2.1 - UML Overview Diagram*



### 3.2.2 Use Case Tables

**Table 1:**

|                      |                                                                  |                                                                              |
|----------------------|------------------------------------------------------------------|------------------------------------------------------------------------------|
| Use Case Name        | <b>Complete Lessons &amp; Tests</b>                              |                                                                              |
| Participating Actors | Learner, AI Engine System                                        |                                                                              |
| Trigger              | 1. Learner clicks start to commence quiz.                        |                                                                              |
| Pre-Conditions       | Users are logged in, and quizzes are available to complete.      |                                                                              |
| Post-Conditions      | The learner receives XP & awards, and the next quiz is unlocked. |                                                                              |
| Success Scenario     |                                                                  | 2. Learner receives quiz questions based on chosen courses and quiz.         |
|                      |                                                                  | 3. Learner inputs/selects answers and submits them.                          |
|                      |                                                                  | 4. System proceeds and presents another question.                            |
|                      |                                                                  | 5. After the learner completes all questions, the system determines results. |

**Table 2:**

|                      |                                                                                            |                                                                      |
|----------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Use Case Name        | <b>Generate Personalized Questions</b>                                                     |                                                                      |
| Participating Actors | Course/Content Managers, AI Engine System                                                  |                                                                      |
| Trigger              | 1. System detects learning patterns requiring customized patterns.                         |                                                                      |
| Pre-Conditions       | Users have learning history, and the AI systems have performance data.                     |                                                                      |
| Post-Conditions      | Learner receives personalized content, and their learning path, and user model is updated. |                                                                      |
| Success Scenario     |                                                                                            | 2. AI analyses user's recent performances to identify learning gaps. |
|                      |                                                                                            | 3. System identifies what content and difficulty the learner needs.  |
|                      |                                                                                            | 4. System generates exercises matching user's needs.                 |
|                      |                                                                                            | 5. System puts personalized content into the user's learning queue.  |
|                      |                                                                                            | 6. User engages with personalized lessons in the next session.       |
|                      |                                                                                            | 7. Systems records data and refines the personalization model.       |

**Table 3:**

|                      |                                                                                                     |                                                                                      |
|----------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Use Case Name        | <b>Manage Subscription</b>                                                                          |                                                                                      |
| Participating Actors | Learner, External System (Payment)                                                                  |                                                                                      |
| Trigger              | 1. User navigates to subscription management system.                                                |                                                                                      |
| Pre-Conditions       | Users have active accounts and can access the payment system.                                       |                                                                                      |
| Post-Conditions      | Payment is processed, subscription is updated and premium features is either activated or disabled. |                                                                                      |
| Success Scenario     |                                                                                                     | 2. The system displays the current subscription status and what plans are available. |
|                      |                                                                                                     | 3. User selects “Subscribe to Duolingo Plus”.                                        |
|                      |                                                                                                     | 4. System processes the payment through the external gateway.                        |
|                      |                                                                                                     | 5. The payment system confirms the transaction.                                      |
|                      |                                                                                                     | 6. System activates premium features.                                                |
|                      |                                                                                                     | 7. System sends confirmation.                                                        |
|                      |                                                                                                     | 8. User gains access to Duolingo Plus.                                               |
| Alternative Flows    | • AF-1: Cancel Subscription                                                                         |                                                                                      |
|                      | At step 3, the user selects “Cancel Subscription”.                                                  |                                                                                      |
|                      | 3.1 System displays cancellation confirmation.                                                      |                                                                                      |
|                      | 3.2 User confirms cancellation.                                                                     |                                                                                      |
|                      | 3.3 System schedules the end of subscription for the current billing period.                        |                                                                                      |
|                      | 3.4 System sets account status to “pending Cancellation.”                                           |                                                                                      |
|                      | 3.5 System sends cancellation confirmation.                                                         |                                                                                      |
|                      | • AF-2: Upgrade Subscription                                                                        |                                                                                      |
|                      | 3.1 System displays upgrade options and pricing.                                                    |                                                                                      |
|                      | 3.2 User selects desired upgrade plan.                                                              |                                                                                      |
|                      | 3.3 System processes upgrade payment.                                                               |                                                                                      |
|                      | 3.4 Payment system confirms transaction is successful.                                              |                                                                                      |
|                      | 3.5 System activates enhanced premium features.                                                     |                                                                                      |
|                      | 3.6 System sends upgrade confirmation.                                                              |                                                                                      |

### 3.2.3 Function Requirements

**Table 1: Complete Lessons and Tests**

- The system shall allow the learner to commence the quiz or test once it is unlocked and available.
- The system shall present quiz or test questions based on chosen course and progress level.
- The system shall accept and record inputs and selected answer from the learner for each question.
- The system shall display a correct or incorrect animation after answer is recorded.
- The system shall automatically proceed and present other questions after inputs have been accepted and recorded.

- The system shall calculate the learner's score and determine whether they passed or not after completing the quiz or test.
  - The system shall award the learner XP, rewards and streaks after passing the lesson or quiz.
  - The system shall unlock the next challenge or quiz after a successful completion.
  - The system shall provide retry options and feedback after an unsuccessful result.
- 
- **Table 2: Generate Personalised Questions**
  - The system shall monitor and analyse completed/recent challenges and quizzes to identify learning trends and knowledge gaps.
  - The system shall identify necessary content difficulty based on the learner's performance metrics
  - The system shall recommend and provide relevant exercises that match the learner's needs.
  - The system shall generate personalized questions and provide it to the learning queue whenever existing material proves inefficient.
  - The system shall record learner's performances when interacting with the personalized content.
  - The system shall refine and update the personalisation model based on the relevant exercise's results.
- 
- **Table 3: Manage Subscription**
  - The system shall present the learner their current subscription status and available subscription plans.
  - The system shall efficiently cooperate with the learner's decisions in subscribing, upgrading plans, free trials and cancellation actions
  - The system shall protect and encrypt learners' personal data when payment details are inputted.
  - The system shall process all payment transactions via the external gateway utilising encryption.
  - The system shall confirm transaction success or failure with the external payment gateway.
  - The system shall set the account to Duolingo Plus and activate its features immediately after confirmation of a successful payment.
  - The system shall disable Duolingo Plus features whenever the subscription expires or is cancelled.
  - The system shall stop the billing cycle whenever the subscription is cancelled.
  - The system shall update and notify the learners frequently via notifications and emails on subscription confirmations, upgrades, updates and cancellations.
  - The system shall ensure secure logging in and payment processes.

### 3.3 Performance Requirements

#### Quality Requirement 1 – Reliability

##### Quality Dimension: Reliability

##### Requirement ID: QR-REL-01

| Attribute             | Content                                                                                                                                                                                                                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description           | The system shall maintain reliable operation during lessons, syncing, collaboration, and partner-media playback without unexpected crashes.                                                                                                                                                                  |
| Rationale             | Learners expect smooth sessions. Teachers need lessons to run during class periods. Kids lose motivation easily when the system pauses or resets.                                                                                                                                                            |
| Satisfaction Criteria | The system shall maintain monthly uptime of $\geq 99.5\%$ .<br>The system shall limit crash rate to under 0.1% per 1,000 active sessions.<br>The system shall restrict downtime to no more than 3 minutes per week.                                                                                          |
| Measurement Method    | Real-time uptime tracking, automated health checks, error-reporting logs, and periodic reliability reviews.                                                                                                                                                                                                  |
| Dependencies          | Reliability has dependence on Duolingo's cloud servers running consistently, the CDN (content delivery network) delivering lesson content quickly and devices having enough memory to avoid abrupt shutdowns. Stable internet is crucial for syncing progress and loading lessons without any interruptions. |
| Risks                 | The risks of reliability with Duolingo's system could be poor Wi-Fi connections, unexpected rises in global activity leading to server crashes, and external API outages which could interrupt certain features.                                                                                             |
| Assumptions           | It's assumed that any user for Duolingo has a basic internet connection, has a device that meets the minimum system requirements, and the cloud provider is continuously online without major incidents.                                                                                                     |

**Quality Requirement 2 – Performance****Quality Dimension: Performance (Speed)****Requirement ID: QR-PERF-01**

| Attribute             | Content                                                                                                                                                                                                                                                                                                               |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description           | The system shall load lessons, dashboards, media suggestions, and teacher reports with minimal delay.                                                                                                                                                                                                                 |
| Rationale             | Slow loading disrupts flow, creates frustration for children, and affects teachers who must move through lessons quickly.                                                                                                                                                                                             |
| Satisfaction Criteria | The system shall load within <b>2 seconds</b> . The system shall respond to API requests within <b>500ms</b> .<br>The system shall load dashboards and teacher views within <b>3 seconds</b> under normal traffic conditions.                                                                                         |
| Measurement Method    | Synthetic load tests, response-time profiling, and continuous performance monitoring tools.                                                                                                                                                                                                                           |
| Dependencies          | Performance for Duolingo depends on swift API responses, an optimised database and a strong content delivery network to delivery to send media assets at a quick rate. Hardware for devices play a role in performance as higher specs like a CPU and RAM are needed to handle the animations or interactive lessons. |
| Risks                 | Slow networks can hinder performance especially in schools and rural areas. Older devices or low-end devices can struggle with animations and high internet traffic speeds which can slow loading times.                                                                                                              |
| Assumptions           | Users are generally expected to have reasonable bandwidth and devices which have specs which can handle Duolingo's features. Backend systems are also assumed to be scaled to an appropriate standard to comply with Duolingo's input.                                                                                |



**Quality Requirement 3 – Security****Quality Dimension: Conformance (to standards)****Requirement ID: QR-SEC-01**

| <b>Attribute</b>      | <b>Content</b>                                                                                                                                                                                                                                                                                                          |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description           | The system shall protect all learner data using strong encryption, controlled access, and regular audits. Child accounts must receive enhanced privacy.                                                                                                                                                                 |
| Rationale             | Parents, schools, and partners expect strong data protection. Security failures damage trust and violate legal requirements.                                                                                                                                                                                            |
| Satisfaction Criteria | The system shall encrypt all data with <b>AES-256</b> at rest and <b>TLS 1.3</b> in transit.<br>The system shall undergo annual penetration testing with all findings addressed.<br>The system shall remediate critical vulnerabilities within <b>14 days</b> .                                                         |
| Measurement Method    | Audit reports, penetration tests, vulnerability scanning, access-log reviews.                                                                                                                                                                                                                                           |
| Dependencies          | Security is relied on encryption methods which keep the users progress safe. Partners like Google and Apple provide accurate access controls for roles like teachers and students. Duolingo's partners are vital for maintaining security standards across the app.                                                     |
| Risks                 | There can be attempts to hack the system like phishing attempts directed at parents and students and finding vulnerable spots in third party APIs. In text-based activities, there can be chances of the user inputting personal information in the activity which can lead to a breach in privacy if a hacking occurs. |
| Assumptions           | Duolingo assumes parents or guardians manage passwords and any other encryption methods, users don't bypass any security settings and partners have an up-to-date authentication system to maintain security standards in the future.                                                                                   |

**Quality Requirement 4 - Usability and Accessibility****Quality Dimension: Usability****Requirement ID: QR-USE-01**

| Attribute             | Content                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description           | The system shall support an intuitive interface, readable content, assistive technologies, and a simplified child mode tailored for early learners.                                                                                                                                                                                                            |
| Rationale             | Duolingo has a wide audience. Children need simple flows. Users with disabilities require accessible formats. Teachers need efficient tools.                                                                                                                                                                                                                   |
| Satisfaction Criteria | The system shall comply with <b>WCAG 2.1 AA</b> accessibility standards.<br>The system shall reduce steps in child mode by <b>30%</b> .<br>The system shall achieve <b>85%+</b> task success in usability tests.                                                                                                                                               |
| Measurement Method    | Accessibility audits, user testing sessions, screen reader evaluations, and log-based task-completion metrics.                                                                                                                                                                                                                                                 |
| Dependencies          | Duolingo's usability depends on a user friendly, intuitive UI with icons that are clear, have accessibility features for students with additional needs, and applicable touch targets to prevent miss clicks. Child mode needs simple navigation for younger users to understand the tasks to do.                                                              |
| Risks                 | There is a risk of younger users getting confused by complex screens or messages, users having disabilities that can struggle to see the UI if accessibility support fails and speaking or listening exercises not working due to the faultiness of the microphones and speakers from the systems or the voice recognition feature unexpectedly stops working. |
| Assumptions           | Users are assumed to have basic mobile features like zoom, contrast and screen reading which aid in their efforts to study with Duolingo.                                                                                                                                                                                                                      |

**Quality Requirement 5 – Scalability****Quality Dimension: Reliability****Requirement ID: QR-SCAL-01**

| <b>Attribute</b>      | <b>Content</b>                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description           | The system shall handle increases in traffic from classroom usage, global challenges, and partner integrations without performance issues.                                                                                                                                                                                                                             |
| Rationale             | Usage patterns spike during school hours or when new features launch. The platform must remain stable under heavy load.                                                                                                                                                                                                                                                |
| Satisfaction Criteria | The system shall support 3× normal load with no more than 20% performance degradation. The system shall auto-scale when usage exceeds normal thresholds for 10 minutes.                                                                                                                                                                                                |
| Measurement Method    | Stress tests, load profiles, cloud scaling logs, and monitoring dashboards.                                                                                                                                                                                                                                                                                            |
| Dependencies          | Duolingo’s scalability is depended on its cloud auto scaling, load balancing across multiple servers and microservices are present to allow the system to horizontally expand during high volumes of usage. Splitting up the databases into “shards” and storing them into different servers (known as database sharding” helps to improve high amounts of user loads. |
| Risks                 | There can be large spikes in internet traffic from classrooms or global use of the app which can lead to excess expected capacity. Partners applications could increase API load which can along with unusual surges of activity can strain infrastructure.                                                                                                            |
| Assumptions           | It’s assumed that the cloud resources can uphold the scalability when triggered and backend systems are designed to handle increases in users and user activity.                                                                                                                                                                                                       |

**Quality Requirement 6 - Compatibility and Integration****Quality Dimension: Compatibility****Requirement ID: QR-COMP-01**

| Attribute             | Content                                                                                                                                                                                                                                                                                                                                     |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description           | The system shall integrate safely with approved third-party platforms, including YouTube Kids, school systems, and NGO learning partners.                                                                                                                                                                                                   |
| Rationale             | Collaboration is one of the system's main goals. Safe integrations expand learning without exposing users to risk.                                                                                                                                                                                                                          |
| Satisfaction Criteria | The system shall have an API success rate $\geq$ <b>99%</b> .<br>The system shall have zero blocked child accounts due to unsafe partner content.<br>The system shall have all integrations reviewed <b>annually</b> .                                                                                                                      |
| Measurement Method    | Integration tests, API error monitoring, partner compliance reports, and periodic audits.                                                                                                                                                                                                                                                   |
| Dependencies          | Compatibility is depended on other partner APIs like YouTube kids remaining stable and users running Duolingo on supported devices and operating systems (e.g. Android 11 or above can only run the latest Duolingo). Up to date API must be documented for adequate integration.                                                           |
| Risks                 | Changes in API partners can cause a risk of incompatible features that could make the app hard to use. Users that rely on older browser versions may not be able to run the best experience of Duolingo. Learning management systems integration with the lessons might fail if there are changes within the external systems data formats. |
| Assumptions           | Duolingo assumes that users update their operating systems to the latest version regularly and the partners document their API endpoints, providing stability.                                                                                                                                                                              |

**Quality Requirement 7 – Maintainability****Quality Dimension: Serviceability****Requirement ID: QR-MAINT-01**

| Attribute             | Content                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description           | The system shall support easy maintenance by using modular components, clean API boundaries, and documented code structures.                                                                                                                                                                                                                                                                                           |
| Rationale             | The platform evolves often. Clear maintenance standards reduce long-term cost and speed up bug fixes.                                                                                                                                                                                                                                                                                                                  |
| Satisfaction Criteria | The system shall use an average time to resolve defects <b>≤ 5 working days</b> .<br>The system shall incorporate code modules documented following internal guidelines.                                                                                                                                                                                                                                               |
| Measurement Method    | Maintenance logs, code review reports, defect resolution metrics.                                                                                                                                                                                                                                                                                                                                                      |
| Dependencies          | Duolingo's maintenance depends on the cleanliness of the code and the efficiency of the automated testing pipelines which help to catch bugs early. Version control systems that give the ability to go back to previous versions can help developers document their changes.                                                                                                                                          |
| Risks                 | Risks can include poorly designed code that could interfere poorly with features, an increase in needless feature additions which can lead to a lot of bugs as there is no proper maintenance and developer turnover which can lead to a reset in principles. Inadequate documentation can also increase the risk of errors during maintenance as there can be errors that don't get enough attention like the others. |
| Assumptions           | It's assumed that the developers abide by coding standards, update documentation regularly, and maintain consistent testing to find bugs.                                                                                                                                                                                                                                                                              |

**Quality Requirement 8 – Portability****Quality Dimension: Portability****Requirement ID: QR-PORT-01**

| Attribute             | Content                                                                                                                                                                                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description           | The system shall operate consistently across supported platforms (Android, iOS, browsers) and adapt to regional restrictions.                                                                                                                         |
| Rationale             | Duolingo supports global learners using different devices and educational systems.                                                                                                                                                                    |
| Satisfaction Criteria | The system shall implement full functionality available on <b>100%</b> of supported platforms with no more than <b>5% variation</b> in performance across devices.                                                                                    |
| Measurement Method    | Cross-platform testing, compatibility reports, device farm automated tests.                                                                                                                                                                           |
| Dependencies          | Portability relies on the use of cross platform frameworks, compatibility maintenance with browsers, and testing on various types of devices with regards to size range and hardware capabilities.                                                    |
| Risks                 | There are risks of Browser updates that can break any UI elements Duolingo may have which could make the application hard to use. Older devices pose a threat to portability as they could not have the resources to run modern elements of Duolingo. |
| Assumptions           | Users are expected to have devices that can run the latest version of Duolingo with their necessary specs without any major issues and update their apps regularly to avoid issues regarding compatibility.                                           |

**Quality Requirement 9 – Efficiency****Quality Dimension: Performance Efficiency****Requirement ID: QR-EFF-01**

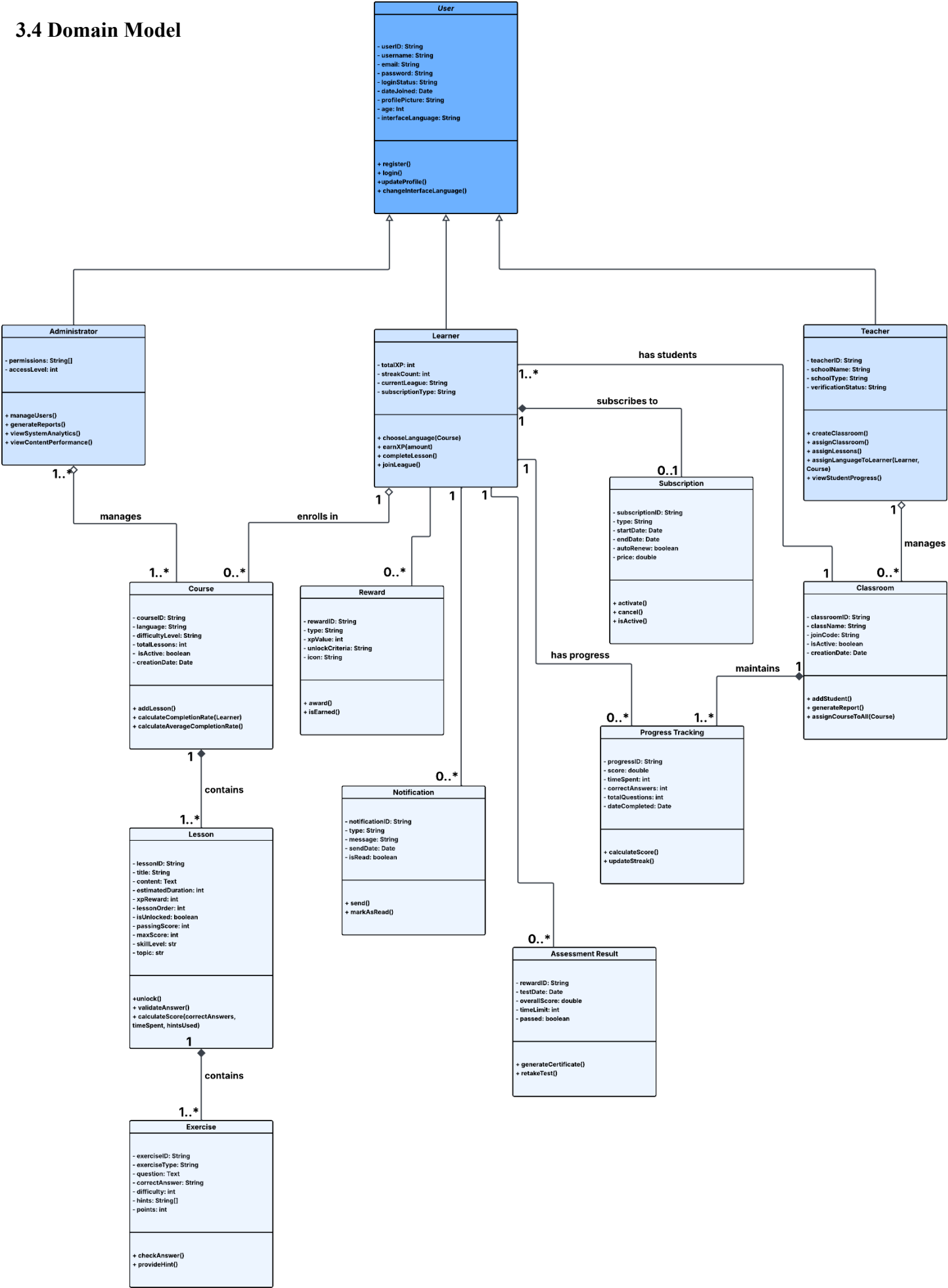
| Attribute             | Content                                                                                                                                                                                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description           | The system shall use processor time, memory, and bandwidth efficiently, so it works on lower-end devices.                                                                                                                                                               |
| Rationale             | Many learners use older phones with limited hardware. Efficient resource use improves accessibility.                                                                                                                                                                    |
| Satisfaction Criteria | The system shall maintain memory usage $\leq$ <b>300MB</b> during lessons.<br>The system shall limit CPU utilisation to $\leq$ <b>40%</b> on mid-tier devices.<br>The system shall have video-based content capped at <b>480p</b> for child mode or low-bandwidth mode. |
| Measurement Method    | Device profiling, mobile performance tests, bandwidth analytics.                                                                                                                                                                                                        |
| Dependencies          | Efficiency is relied on the optimisation of media files, memory usage that is efficient and hardware acceleration for animations. Good caching of unnecessary data can help speed up the apps process and the repeated interactions.                                    |
| Risks                 | Older devices with low RAM may run out of storage quick, causing slow speeds and eventually crashes. High data consumption affects users with limited plans and inefficient code can lead to battery drainage due to the lack of efficiency.                            |
| Assumptions           | User's devices are assumed to have the adequate amount of RAM and processing power to be able to access and allow enough power for typical lesson use, along with users allowing caching for supporting offline and low bandwidth modes.                                |

**Quality Requirement 10 – Flexibility****Quality Dimension: Maintainability****Requirement ID: QR-FLEX-01**

| Attribute             | Content                                                                                                                                                                                                                                                                                         |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description           | The system shall allow new features (e.g., new languages, parental tools, class dashboards, media integrations) to be added without major redesign.                                                                                                                                             |
| Rationale             | Duolingo updates frequently. The ecosystem includes children, schools, and partners, so flexibility supports future growth.                                                                                                                                                                     |
| Satisfaction Criteria | The system shall allow adding a new language or feature requires modifications to <b>no more than 3 core modules</b> .                                                                                                                                                                          |
| Measurement Method    | Architecture review, dependency analysis, change impact logs.                                                                                                                                                                                                                                   |
| Dependencies          | The system shall depend on modular architecture.                                                                                                                                                                                                                                                |
| Risks                 | New features might become too complex for the system to handle smoothly. Over time different parts of the app could become too connected making it harder to add new languages or tools. Sometimes a simple-looking new feature might unexpectedly require changes to core parts of the system. |
| Assumptions           | It's assumed that new features will generally fit within Duolingo's current design approach. Development teams will continue to follow good design practices to keep the system flexible. Future updates will not require a complete rebuild of the platform.                                   |



3.4 Domain Model



### 3.4.1 Key Associations

1. **User Generalization** -> Learner, Teacher, Administrator
2. **Learner** enrolls in **Course**
3. **Learner** completes **Lesson**
4. **Learner** subscribes to **Subscription**
5. **Course** contains **Lesson**
6. **Lesson** contains **Exercise**
7. **Teacher** manages **Classroom**
8. **Classroom** has **Learner**
9. **Classroom** has assigned **Courses**
10. **Learner** has **Progress Tracking**
11. **Lesson** has **skill Level**
12. **Learner** receives **rewards**

### 3.4.2 Role Names

**Learner** - Someone who does exercises to learn

**Course** - Different languages

**Lesson** - Contains exercises that can be of different languages

**Exercise** - Questions learners Answer

**Paid User** - They have an active Subscription Plan

**Progress Tracking** - Track progress

**Teacher** - Manages a classroom

**Classroom** - Group of Students

**Administrator** - Admin of the app

**Notification** - Notify users

**Assessment Result** - Results from lessons

### 3.4.3 Cardinalities/Multiplicities

- (1)Learner subscribes to (0..1) Subscription
- (1..\*)Administrator manages (1..\*) Course(s)
- (1)Learner enrolls in (0..\*) course
- (1)Learner receives (0..\*) notification(s)
- (1)Learner receives (0..\*) reward(s)
- (1)Learner receives (0..\*) assessment result(s)

- (1)Teacher manages (0..\*) classroom(s)
- (1)Classroom has (1..\*) student(s)
- (1)Classroom maintains (1..\*) progress tracker(s)
- (1)course Contains (1..\*) lesson(s)
- (1)Lesson contains (1..\*) exercise(s)

#### 3.4.4 Attributes and Operations:

- User: userId:str, username:str, email:str, password:str, loginStatus:str, dateJoined:date, profilePicture:str, age:int, interfaceLanguage:str, register(), login(), updateProfile(), changeinterfaceLanguage()
- Administrator: permissions:str, accessLevel:int, manageUsers(), generateReports(), viewSystemAnalytics(), viewContentPerformance()
- Learner: totalXp:int, streakCount:int, currentLeague:str, subscriptionType:str, chooseLanguage(Course), earnXp(amount), completeLesson(), joinLeague()
- Teacher: teacherId:str, schoolName:str, schoolType:str, verificationStatus:str, createClassroom(), assignClassroom(), assignLessons(), assignLanguageToLearner(Learner,Course), viewStudentProgress()
- Course: courseId:str, language:str, difficultyLevel:str, totalLessons:int, isActive:bool, creationDate:date, addLesson(), calculateCompletionRate(Learner), calculateAverageCompletionRate()
- Lesson: lessonId:str, title:str, content:text, estimatedDuration:int, xpReward:int, lessonOrder:int, isUnlocked:bool, passingScore:int, maxScore:int, skillLevel: str, topic: str, unlock(), validateAnswer(), calculateScore(correctAnswers,timeSpent,hintsUsed)
- Exercise: exerciseId:str, exerciseType:str, question:text, correctAnswer:str, difficulty:int, hints:str, points:int, checkAnswer(), provideHint()
- Reward: rewardId:str, type:str, xpValue:int, unlockCriteria:str, icon:str, award(), isEarned()
- Notification: notificationsId:str, type:str, message:str, sendData:date, isRead:bool, send(), markAsRead()
- Assessment result: rewardId:str, testDate:date, overallScore:double, timeLimit:int, passed:bool, generateCertificate(), retakeTest()
- Subscription: subscriptionId:str, type:str, startDate:date, endDate:date, autoRenew:bool, price:double, activate(), cancel(), isActive()
- Progress Tracking: progressID:str, score:double, timeSpent:int, correctAnswers:int, totalQuestions:int, dateCompleted:date, calculateScore(), updateStreak()
- Classroom: classroomId:str, className:str, joinCode:str, isActive:bool, creationDate:date, addStudent(), generateReport(), assignCourseToAll(course)

## **4. Specific additional Requirements**

This section provides an in-depth and precise description of how the Duolingo system must behave, perform, interact, and operate. It translates the earlier goals and vision into concrete, testable, and implementation-ready requirements. Each subsection is structured to ensure that a developer, tester, or system architect can understand exactly what is expected without needing assumptions or external clarification.

All requirements below are written to be:

- Clear and unambiguous
- Measurable and verifiable
- Directly linked to user needs and system goals
- Suitable for both implementation and evaluation

### **4.1 External Interface Requirements**

This section defines how the Duolingo system communicates with all external elements, including users, devices, third-party software, and communication networks.

#### **4.1.1 User Interface**

The Duolingo system shall provide a user interface that supports clarity, consistency, accessibility, and ease of use for all user groups, including children, adults, teachers, and parents.

The user interface shall:

- Present a clean and distraction-free layout focused on learning tasks.
- Use large, readable fonts suitable for children and visually impaired users.
- Display clear visual indicators for progress, such as progress bars, XP counters, and streak icons.
- Use consistent iconography across all screens (home, lesson, profile, leaderboard, settings).
- Provide intuitive navigation with a maximum of 3–5 taps to reach core features.

Specific interface elements shall include:

- A learner dashboard displaying:
  - Current level
  - Daily goal progress
  - Streak status
  - Recommended next lesson
- A child mode interface featuring:
  - Simplified navigation
  - Bright, friendly visuals
  - Minimal text complexity
  - Extra confirmation steps before exiting lessons
- A teacher dashboard with:
  - Class overview panel
  - Student performance table

- Assignment creation interface
- Progress comparison tools
- A parent dashboard providing:
  - Screen time limits
  - Weekly performance summary
  - Content control settings
  - Safety alerts

The system shall also:

- Support light and dark themes.
- Provide on-screen tooltips for first-time users.
- Allow interface language switching at any time.
- Support screen readers and voice navigation tools.

The interface design must reduce cognitive load and ensure smooth task completion.

### **4.1.2 Hardware Interfaces**

The system interacts with device hardware through standard input and output components. These interactions must support learning quality without requiring specialised equipment.

Supported hardware interactions include:

- Touchscreen input for navigation and lesson interaction.
- Microphone input for pronunciation exercises and speech recognition.
- Speakers or headphones for listening comprehension activities.
- Camera access for profile setup or identity verification if required.
- Haptic feedback (vibration) for task completion confirmation.

The system shall:

- Detect unavailable hardware automatically.
- Display user-friendly alerts when hardware access is denied.
- Allow users to skip hardware-dependent tasks if needed.
- Adjust lesson modes when audio or visual input is restricted.

No additional external hardware is required beyond standard smartphone or computer components.

### **4.1.3 Software Interfaces**

The Duolingo system shall communicate with several internal and external software systems using secure and standardised interfaces.

Internal software systems include:

- User authentication system
- Learning progress tracking system
- Lesson content management system
- Analytics processing engine
- Notification dispatch service

External software systems include:

- YouTube Kids for embedded educational videos
- Google Firebase and AWS for backend services
- School Learning Management Systems (LMS)
- Payment service providers
- App store services (Google Play, Apple App Store)

The system shall:

- Use RESTful APIs for data exchange.
- Use HTTPS with token-based authentication for security.
- Log all integration activities for audit purposes.
- Gracefully handle interface failure without system crash.

## 4.1.4 Communication Interfaces

The system shall use secure and reliable communication channels to transfer data.

Communication protocols include:

- HTTPS for data transmission
- Push notification services
- Encrypted messaging protocols

The system shall ensure:

- End-to-end encryption of sensitive data
- Low latency for time-sensitive features
- Stable connectivity handling during bandwidth fluctuations

## 4.2 Functional Requirements

This section defines what the Duolingo system must do to achieve its core purpose.

### 4.2.1 Learning Functionality

The system shall provide comprehensive language learning tools that adapt to user ability and learning pace.

It shall:

- Deliver structured lessons based on user proficiency level.
- Automatically adjust difficulty levels based on performance trends.
- Provide real-time feedback for answers.
- Reinforce learning through spaced repetition.
- Support pronunciation analysis and correction.

The system must track:

- XP points
- Daily streaks
- Skill progression
- Common error patterns

It shall motivate users using:

- Achievement badges
- Milestone rewards
- Encouraging feedback visuals

#### **4.2.2 Collaboration Functionality**

The system shall support collaboration between learners, teachers, and institutions.

It shall allow:

- Formation of class groups
- Teacher-assigned tasks
- Group challenges
- Peer interaction and feedback

Teachers shall be able to:

- Track student performance
- Send feedback and instructions
- Adjust lesson difficulty

#### **4.2.3 Parental Controls**

The system shall provide robust parental oversight tools.

Parents shall:

- View learning activity logs
- Set screen time restrictions
- Receive alerts for unusual activity
- Restrict content types

Child accounts shall include:

- Age-appropriate content filters
- Restricted chat capabilities
- Safe browsing features

#### **4.2.4 Analytics and Reporting**

The system shall provide detailed and visual reports.

It shall generate:

- Daily progress summaries
- Weekly performance analytics
- Class performance comparisons
- Error frequency analysis

These reports shall help identify learning patterns and improvement areas.

#### **4.2.5 Partner Integrations**

The system shall integrate external educational resources.

It shall:

- Display embedded video lessons
- Provide verified content links
- Track partner content interaction
- Control partner content based on age settings

## **4.3 Performance Requirements**

### **4.3.1 Response Time**

The system must respond efficiently to maintain user focus.

Targets:

- Lesson load time  $\leq 2$  seconds
- Interactive response time  $\leq 1$  second
- Report generation  $\leq 3$  seconds

### **4.3.2 Scalability**

The system shall scale to support millions of users while maintaining performance integrity.

It shall support:

- Peak traffic loads
- Class-based bulk sessions
- Seasonal demand increases

### **4.3.3 Reliability Targets**

System uptime shall not fall below 99.5% monthly.

Maintenance windows shall be communicated in advance.

## **4.4 Logical Database Requirements**

### **4.4.1 Data Entities**

Key data entities include:

- User
- Lesson
- Skill
- Progress
- Achievement
- Parent
- Teacher
- Class
- Content Module

### **4.4.2 Relationship and Constraints**



Key relationships include:

- One user - > many lessons
- One teacher - > many classes
- One parent - > many children
- Every class - > exactly one teacher

Constraints ensure data integrity and accountability.

#### **4.4.3 Data Flow Diagram**

User Input → Lesson Engine → Progress Processor → Database → Analytics Processor → Dashboard  
Display → Feedback Loop

#### **4.5 Design Constraints**

##### **4.5.1 Standards Compliance**

System must comply with GDPR, COPPA, FERPA, and WCAG accessibility standards.

##### **4.5.2 Development Environment**

The system shall use:

- Continuous integration pipelines
- Cloud hosting platforms
- Version control systems

##### **4.5.3 Programming Languages and Tools**

Primary tools include:

- Python
- JavaScript
- React
- Node.js
- AWS services

#### **4.6 Software System Attributes**

Security, usability, maintainability, scalability, availability, and portability shall be monitored continuously.

These attributes ensure long-term sustainability and user trust.

#### **4.7 Organising the specific Requirements**

##### **4.7.1 Requirements Prioritization**

High:

- Authentication
- Lesson delivery
- Parental controls

Medium:

- Partner integrations
- Advanced reporting

Low:

- Visual customization features

#### **4.7.2 Requirement Traceability Matrix**

Requirement to Goal Mapping

- Adaptive lessons - > Engagement
- Child mode - > Safety
- External integration - > Collaboration

#### **4.8 Additional Comments**

##### **4.8.1 Future Work**

Future improvements may include:

- AI-guided tutoring
- VR-based language learning
- Enhanced offline capabilities

##### **4.8.2 Known Risks**

- Partner system instability
- Data breaches
- High server load during peak periods

##### **4.8.3 Assumptions and Limitations**

Assumptions:

- Continuous internet access
- Stable partner services

Limitations:

- Offline access limited
- Some features restricted to premium users